Hints: Integrals – 5. The gamma and beta functions and Wallis' formulas

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Substitute $u = ax^b$: $x = (u/a)^{1/b}$, $dx = \frac{1}{b}a^{-1/b}u^{1/b-1}du$, and $x^{s-1}dx = \frac{1}{b}a^{-s/b}u^{s/b-1}du$.

Hint 2. $\int_0^\infty e^{-x^3}dx = \frac{1}{3}\Gamma(\frac{1}{3}) = \Gamma(\frac{4}{3})$; $\int_0^\infty x^5e^{-x^4}dx = \frac{1}{4}\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{8}$; $\int_0^\infty x^3e^{-x^2}dx = \frac{1}{2}\Gamma(2) = \frac{1}{2}$.

Problem 2

Hint 1. With $x = \sin^2 t$: $B(s,s) = 2\int_0^{\pi/2}(\sin t\cos t)^{2s-1}dt = 2^{2-2s}\int_0^{\pi/2}\sin^{2s-1}2t\,dt$.

Hint 2. Substituting $u = 2t$ and using symmetry about $\frac{\pi}{2}$, the last integral is $\int_0^{\pi/2}\sin^{2s-1}u\,du = \frac{1}{2}B(s,\frac{1}{2})$. So $B(s,s) = 2^{1-2s}B(s,\frac{1}{2})$, i.e. $\frac{\Gamma(s)^2}{\Gamma(2s)} = 2^{1-2s}\frac{\Gamma(s)\sqrt{\pi}}{\Gamma(s+\frac{1}{2})}$, which is the duplication formula.

Problem 3

Hint 1. Substitute $u = x^n$: the integral becomes $\frac{1}{n}\int_0^1 u^{1/n-1}(1-u)^{-1/n}du = \frac{1}{n}B\left(\frac{1}{n}, 1-\frac{1}{n}\right)$.

Hint 2. By the reflection formula this is $\frac{\pi/n}{\sin(\pi/n)}$, which tends to 1 as $n\to\infty$.

Problem 4

Hint 1. Wallis' product: as in the class sheet, squeeze I_{2m} between I_{2m+1} and I_{2m-1}, $I_n = \int_0^{\pi/2}\sin^n x\,dx$. An equivalent form: $\frac{(2m)!!}{(2m-1)!!} = \frac{4^m(m!)^2}{(2m)!} \sim \sqrt{\pi m}$.

Hint 2. Stirling up to a constant: let $a_n = \frac{n!\,e^n}{n^{n+1/2}}$. Then $\ln\frac{a_n}{a_{n+1}} = (n+\frac{1}{2})\ln(1+\frac{1}{n}) - 1$, which is positive and $O(1/n^2)$; so a_n decreases to a limit $C > 0$ (show it stays positive by summing the $O(1/n^2)$ bounds).

Hint 3. Insert $n! \sim Cn^{n+1/2}e^{-n}$ into $\frac{4^m(m!)^2}{(2m)!}$: it becomes $\sim \frac{C\sqrt{m}}{\sqrt{2}}$. Comparing with $\sqrt{\pi m}$ gives $C = \sqrt{2\pi}$.

Problem 5

Hint 1. $\tan^a x = \sin^a x\cos^{-a}x$, so by Fact 1 of the class sheet the integral is $\frac{1}{2}B\left(\frac{1+a}{2}, \frac{1-a}{2}\right) = \frac{1}{2}\Gamma\left(\frac{1+a}{2}\right)\Gamma\left(\frac{1-a}{2}\right)$.

Hint 2. By reflection it equals $\frac{\pi}{2\sin\frac{\pi(1+a)}{2}} = \frac{\pi}{2\cos\frac{\pi a}{2}}$, which is even in a. Directly: $x \mapsto \frac{\pi}{2} - x$ turns $\tan^a x$ into $\cot^a x = \tan^{-a}x$.

Problem 6

Hint 1. Substitute $x = e^{-u}$, so $\ln\frac{1}{x} = u$ and $dx = -e^{-u}du$.

Hint 2. You get $\int_0^\infty u^{-1/2}e^{-u}du = \Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\int_0^\infty u^{s-1}e^{-u}du = \Gamma(s)$.

Problem 7

Hint 1. Log-convexity: for $\lambda \in (0,1)$ write $x^{\lambda s+(1-\lambda)t-1}e^{-x} = \left(x^{s-1}e^{-x}\right)^{\lambda}\left(x^{t-1}e^{-x}\right)^{1-\lambda}$ and apply Hölder with exponents $\frac{1}{\lambda}$, $\frac{1}{1-\lambda}$: $\Gamma(\lambda s+(1-\lambda)t) \leqslant \Gamma(s)^{\lambda}\Gamma(t)^{1-\lambda}$.

Hint 2. Bohr–Mollerup: if $f\colon (0,\infty)\to(0,\infty)$ satisfies $f(1)=1$, $f(s+1)=sf(s)$ and $\ln f$ is convex, then $f=\Gamma$. Idea of proof: for $0<x\leqslant 1$, compare slopes of $\ln f$ on $[n,n+1]$, $[n+1,n+1+x]$, $[n+1,n+2]$ to get $n^x \leqslant \frac{f(n+1+x)}{n!} \leqslant (n+1)^x$; with the functional equation this forces $f(x)=\lim_{n\to\infty}\frac{n!\,n^x}{x(x+1)\cdots(x+n)}$, a formula that does not depend on f.

Problem 8

Hint 1. Expand $\frac{1}{e^x\pm 1} = \sum_{k\geqslant 1}(\mp 1)^{k-1}e^{-kx}$: then $\int_0^\infty \frac{x^{s-1}}{e^x+1}dx = \Gamma(s)\eta(s)$ and $\int_0^\infty \frac{x^{s-1}}{e^x-1}dx = \Gamma(s)\zeta(s)$. With $s=1$ resp. $s=2$: $\eta(1)=\ln 2$ and $\Gamma(2)\zeta(2)=\frac{\pi^2}{6}$ (the first is also immediate: $\int_0^\infty \frac{e^{-x}}{1+e^{-x}}dx = \ln 2$).

Hint 2. For the third, expanding $\frac{1}{1+e^{2x}}$ gives Gaussians with shifted centres and an alternating series that does not lead anywhere simple. Instead, $x\mapsto -x$ turns $\frac{1}{1+e^{2x}}$ into $\frac{e^{2x}}{1+e^{2x}}$ while e^{-x^2} is even; adding, $2I=\int_{-\infty}^{\infty}e^{-x^2}dx$, so $I=\frac{\sqrt{\pi}}{2}$.

Problem 9

Hint 1. Compute $\int_{\mathbb{R}^n} e^{-|x|^2}dx$ in two ways: as $\left(\int e^{-t^2}dt\right)^n = \pi^{n/2}$, and in spherical shells, $\int_0^\infty e^{-r^2}\,nV_n r^{n-1}dr$, where $nV_n r^{n-1}$ is the area of the sphere of radius r.

Hint 2. The radial integral is $\frac{nV_n}{2}\Gamma(\frac{n}{2}) = V_n\Gamma(\frac{n}{2}+1)$, so $V_n=\frac{\pi^{n/2}}{\Gamma(\frac{n}{2}+1)}$: $V_2=\pi$, $V_3=\frac{4}{3}\pi$, $V_4=\frac{\pi^2}{2}$.

Hint 3. By Stirling, $\Gamma(\frac{n}{2}+1)\sim\sqrt{\pi n}\,(\frac{n}{2e})^{n/2}$, so $V_n\sim\frac{1}{\sqrt{\pi n}}\left(\frac{2\pi e}{n}\right)^{n/2}\to 0$ faster than any exponential.

Problem 10

Hint 1. Use $\frac{1}{x^s}=\frac{1}{\Gamma(s)}\int_0^\infty t^{s-1}e^{-xt}dt$ and swap the integrals; the inner integral is $\int_0^\infty e^{-xt}\sin x\,dx=\frac{1}{1+t^2}$.

Hint 2. You get $\frac{1}{\Gamma(s)}\int_0^\infty\frac{t^{s-1}}{1+t^2}dt$, and $u=t^2$ turns this into $\frac{1}{2\Gamma(s)}\int_0^\infty\frac{u^{s/2-1}}{1+u}du=\frac{\pi}{2\Gamma(s)\sin\frac{\pi s}{2}}$ (class sheet, Problem 4). At $s=1$: $\frac{\pi}{2}$.

Hint 3. The swap needs care: the double integral is not absolutely convergent near $x=\infty$. Insert $e^{-\varepsilon x}$ (then Fubini applies, with inner integral $\frac{1}{1+(t+\varepsilon)^2}$) and let $\varepsilon\to 0^+$, using dominated convergence in t and Abel-type uniform convergence in x as in the $\frac{\sin x}{x}$ computation.